

Name	Puteri Binti Ainul Hasni
Student ID	36853577
Studio Session	14

Part A: URL of the Visualisation

<https://pbin0009.github.io/Data-Visualisation-2/>

Part B: Domain Description

Chart 1: Fast Food Restaurant Around Australia (Dot Map)

Domain Description

This visualization belongs to the public health and food accessibility domain. It examines the geographic distribution of fast-food restaurants across Australia and their potential relationship with population density and obesity trends.

Why:

The map aims to show where fast-food restaurants are concentrated across Australia. Understanding restaurant distribution can help identify areas with greater access to fast food and support discussions about dietary habits and health outcomes.

Who:

The target audience includes health researchers, policymakers, students, and the general public interested in food accessibility and public health issues.

What

The visualization uses spatial data containing the geographic coordinates (latitude and longitude) of fast-food restaurant locations across Australia.

Attributes:

- Restaurant location (latitude and longitude)
- State/territory boundaries

Data Types:

- Spatial (location coordinates)
- Nominal (restaurant category)

How

A **dot map** is used to represent individual fast-food restaurant locations.

Marks:

- Points

Channels:

- Position (longitude and latitude)
- Color (orange points representing restaurants)

The dot map was chosen because it effectively displays the distribution and clustering of restaurants. Users can easily observe that restaurants are highly concentrated in major metropolitan and coastal regions such as Sydney, Melbourne, and Brisbane.

Chart 2: Fast Food Type Market Share (Donut Chart)

Domain Description

This visualization belongs to the food industry and consumer behaviour domain. It explores the market share of different fast-food categories within Australia's fast-food sector.

Why:

The chart aims to compare the popularity and market share of different fast-food categories. It helps users understand which types of fast food dominate the Australian market.

Who:

The target audience includes consumers, researchers, students, and policymakers interested in food consumption patterns and industry trends.

What

The visualization uses market share data representing the percentage contribution of different fast-food categories.

Attributes:

- Fast food type
- Market share percentage

Data Types:

- Nominal (fast food category)
- Quantitative (market share percentage)

How

A **donut chart** is used to show the proportion of each fast-food category relative to the whole market.

Marks:

- Arc segments

Channels:

- Angle (market share percentage)
- Color (different food categories)

The donut chart was selected because it allows viewers to quickly compare category proportions and identify dominant categories. The visualization clearly shows that burgers/sandwiches and chicken contribute the largest share of Australia's fast-food market.

Chart 3: Average Monthly Fast Food Spending by State (Lollipop Chart)

Domain Description

This visualization belongs to the public health and consumer spending domain. It examines the average amount spent on fast food by consumers across different Australian states.

Why:

The chart aims to compare fast-food spending levels between states and identify regions where consumers spend more on fast food. This may provide insights into consumption patterns that could be associated with dietary behaviours.

Who:

The target audience includes policymakers, health researchers, students, and the general public interested in consumer spending and food consumption trends.

What

The visualization uses data on average monthly fast-food expenditure for different Australian states.

Attributes:

- State
- Average monthly fast-food spending (AUD)

Data Types:

- Nominal (State)
- Quantitative (Average spending)

How

A **lollipop chart** is used to compare spending values across states.

Marks:

- Lines
- Points

Channels:

- Position on the x-axis (spending amount)
- Position on the y-axis (state)
- Length of line (magnitude of spending)

The lollipop chart was selected because it provides a cleaner alternative to a bar chart while making comparisons between states easy. The chart shows that New South Wales and Victoria have the highest average monthly fast-food spending, while Tasmania has the lowest.

Chart 4: QSR Consumer Habits in Australia (Horizontal Bar Chart)

Domain Description

This visualization belongs to the consumer behaviour and food industry domain. It explores common habits and preferences of Australian consumers when purchasing quick-service restaurant (QSR) food.

Why:

The chart aims to identify the most common behaviours associated with fast-food consumption. Understanding these habits helps explain consumer preferences and purchasing decisions.

Who:

The target audience includes researchers, policymakers, businesses, students, and consumers interested in food consumption trends and customer behaviour.

What

The visualization uses survey-based data showing the percentage of consumers engaging in various fast-food-related behaviours.

Attributes:

- Consumer habit
- Percentage of respondents

Data Types:

- Nominal (consumer habit category)
- Quantitative (percentage)

How

A horizontal bar chart is used to compare the percentage of consumers exhibiting different behaviours.

Marks:

- Bars

Channels:

- Length of bar (percentage value)
- Position on the y-axis (habit category)
- Position on the x-axis (percentage)

The horizontal bar chart was chosen because it allows easy comparison of multiple categories with long text labels. The visualization shows that the most common behaviour is ordering chips and a drink with a meal, while carrying a QSR membership is among the least common habits.

Chart 5 & 6 : Severe Obesity in Adults Over Time & Child and Adolescent Obesity Trends Over Time

Domain Description

These visualizations belong to the public health domain and focus on obesity trends in Australia. They examine how obesity rates have changed over time among different population groups, including adults, children, and adolescents.

Why:

The line charts aim to reveal long-term obesity trends and allow users to compare changes across demographic groups. Understanding these trends can help identify whether obesity has become more prevalent over time and support discussions about public health challenges and prevention strategies.

Who:

The target audience includes policymakers, health researchers, students, healthcare professionals, and members of the public interested in obesity and population health in Australia.

What

The visualizations use obesity data collected across multiple years in Australia.

Attributes:

- Year
- Obesity rate (%)
- Gender (Male, Female)
- Weight status (Obese, Overweight)

Data Types:

- Temporal (Year)
- Quantitative (Obesity rate)
- Nominal (Gender, Weight status category)

How

Both visualizations use line charts to display changes in obesity rates over time.

Marks:

- Lines
- Points

Channels:

- Position on the x-axis (Year)
- Position on the y-axis (Obesity rate)
- Color (Different groups such as Male/Female and Obese/Overweight)

Line charts were chosen because they effectively show trends and changes over time. The charts allow users to easily observe increases, decreases, and fluctuations in obesity rates across different demographic groups. The visualizations show that obesity rates have generally increased over the years, with adult obesity rising steadily and child and adolescent obesity remaining relatively stable with minor fluctuations.

Chart 7: Overweight and Obesity Rates Across Age Groups (Dumbbell Chart)

Domain Description

This visualization belongs to the public health domain and shows overweight and obesity rates across different age groups in Australia.

Why:

To compare overweight and obesity rates between age groups and identify how prevalence changes with age.

Who:

Policymakers, health researchers, healthcare professionals, students, and the general public.

What

The visualization uses obesity and overweight prevalence data grouped by age category.

Attributes:

- Age group
- Overweight rate (%)
- Obesity rate (%)

Data Types:

- Ordinal (Age group)
- Quantitative (Percentage)
- Nominal (Weight status)

How

A **dumbbell chart** is used to compare overweight and obesity rates within each age group.

Marks:

- Points
- Lines

Channels:

- Position (percentage and age group)
- Color (overweight vs obesity)

The dumbbell chart clearly highlights the gap between overweight and obesity rates across age groups and makes comparisons easy to understand.

Chart 8 : BMI Classification Aged 18 and Over by Gender in 2022 (Stacked Bar Chart)

Domain Description

This visualization belongs to the public health domain and shows the distribution of BMI classifications among Australian adults by gender in 2022.

Why:

To compare BMI classifications between males and females and identify differences in weight status.

Who:

Policymakers, health researchers, healthcare professionals, students, and the general public.

What

Attributes:

- Gender
- BMI classification
- Percentage of population

Data Types:

- Nominal (Gender, BMI classification)
- Quantitative (Percentage)

How

A **stacked bar chart** is used to show the proportion of each BMI category within each gender.

Marks:

- Bars

Channels:

- Length (percentage)
- Color (BMI category)
- Position (gender)

The stacked bar chart allows easy comparison of BMI category distributions between males and females.

Chart 9 : BMI Distribution of Persons Aged 18 and Over in 2022 (Heatmap)

Domain Description

This visualization belongs to the public health domain and shows the distribution of BMI ranges among Australian adults by gender in 2022.

Why:

To identify which BMI ranges contain the largest proportion of males and females.

Who:

Policymakers, health researchers, healthcare professionals, students, and the general public.

What

Attributes:

- Gender
- BMI range
- Percentage of population

Data Types:

- Nominal (Gender)
- Ordinal (BMI range)
- Quantitative (Percentage)

How

A **heatmap** is used to display BMI distribution across different BMI ranges.

Marks:

- Rectangles (cells)

Channels:

- Position (gender and BMI range)
- Color intensity (percentage)

The heatmap helps users quickly identify BMI ranges with higher population concentrations through variations in color intensity.

Chart 10 : Insufficient Weekly Physical Activity of Adults by PHN Area (Choropleth Map)

Domain Description

This visualization belongs to the public health domain and shows the percentage of adults with insufficient physical activity across PHN regions in Australia.

Why:

To identify regional differences in physical inactivity and highlight areas with higher inactivity rates.

Who:

Policymakers, health researchers, healthcare professionals, and the general public.

What

Attributes:

- PHN area
- Percentage of inactive adults

Data Types:

- Spatial (PHN region)
- Quantitative (Percentage)

How

A **choropleth map** is used to display inactivity rates across regions.

Marks:

- Areas (PHN regions)

Channels:

- Position (geographic location)
- Color intensity (inactivity percentage)

The choropleth map allows users to quickly compare physical inactivity levels across different regions of Australia.

Chart 11 : Weekly Physical Activity Duration by Gender (2022) (Butterfly Bar Chart)

Domain Description

This visualization belongs to the public health domain and shows the distribution of weekly physical activity duration among Australian adults by gender in 2022.

Why:

To compare physical activity duration patterns between males and females.

Who:

Policymakers, health researchers, healthcare professionals, students, and the general public.

What

Attributes:

- Gender
- Physical activity duration category
- Percentage of population

Data Types:

- Nominal (Gender)
- Ordinal (Duration category)
- Quantitative (Percentage)

How

A butterfly bar chart is used to compare weekly physical activity duration between males and females.

Marks:

- Bars

Channels:

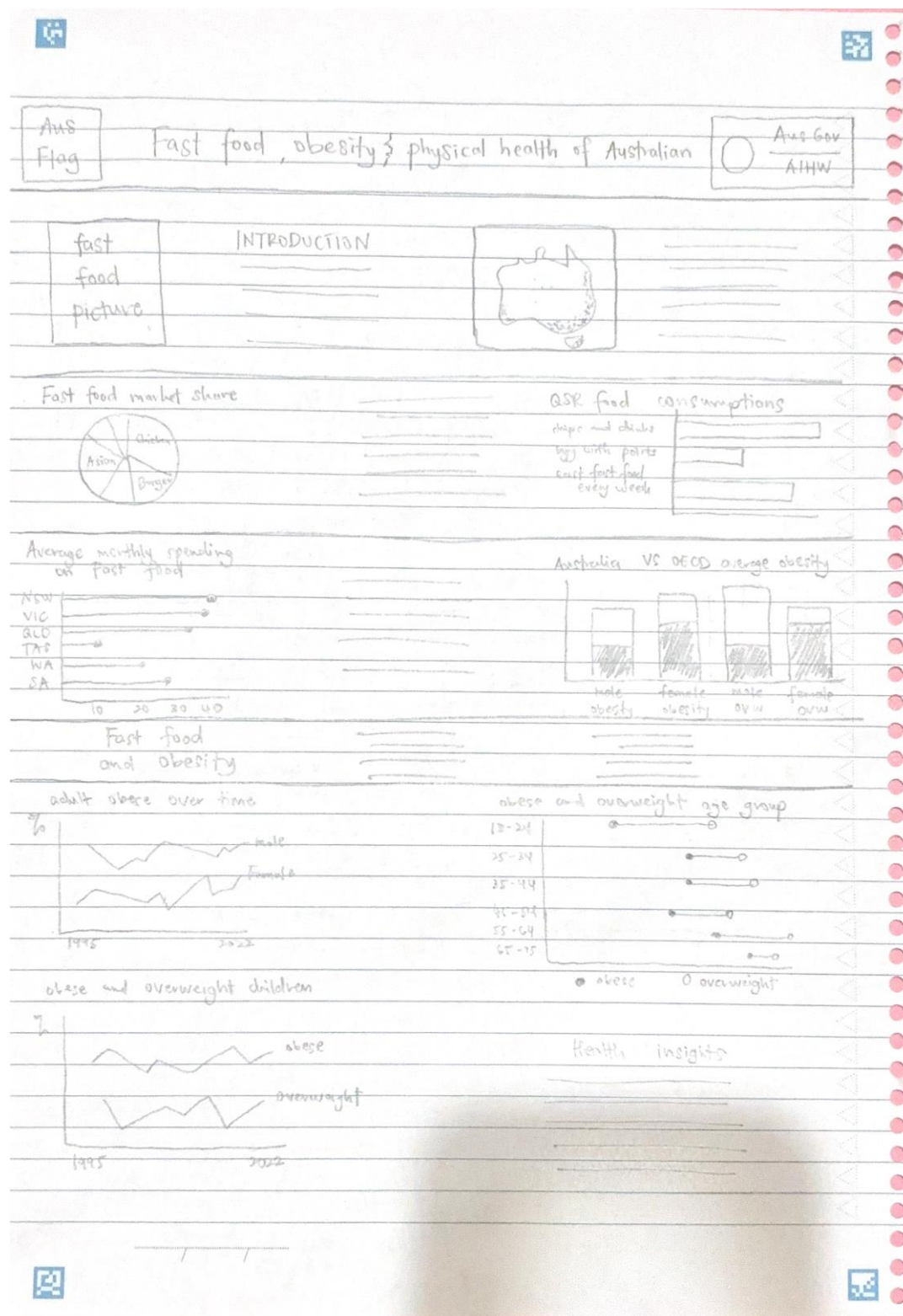
- Length (percentage)
- Position (duration category)
- Color (gender)

The butterfly bar chart allows direct comparison between male and female participation across different physical activity duration categories by placing the two groups on opposite sides of a central axis.

Part C: Sketch

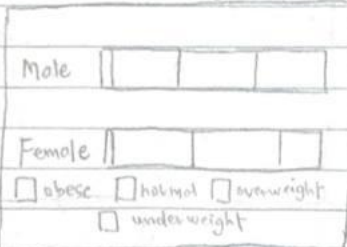
[https://github.com/pbin0009/Data-Visualisation-](https://github.com/pbin0009/Data-Visualisation-2/blob/191939ce0cebd6cdd199eaaf5ff0b84c35601e14/DV2%20Sketch.pdf)

[2/blob/191939ce0cebd6cdd199eaaf5ff0b84c35601e14/DV2%20Sketch.pdf](https://github.com/pbin0009/Data-Visualisation-2/blob/191939ce0cebd6cdd199eaaf5ff0b84c35601e14/DV2%20Sketch.pdf)



BMI and obesity

BMI index adult by Gender



BMI distribution of Adult Australians 2022

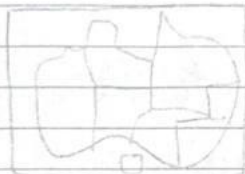
BMI Range	Male (%)	Female (%)
18.5 - 20.5	1.6	2.3
20.5 - 22.5	7.2	6.6
22.5 - 24.5	6.8	11.2
24.5 - 26.5	12.4	15.1
26.5 - 28.5	14.2	12.7
28.5 - 30.5	11.3	11.9
30.5 - 32.5	13.1	10.8
32.5 - 34.5	10.5	8.2
34.5 - 36.5	6.8	5.5
36.5 - 38.5	4.3	4.7

Percentage %

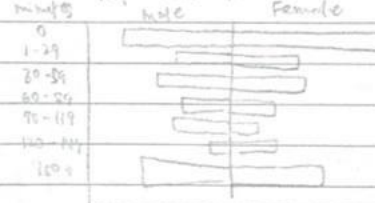
male female

obesity and physical activity

Adults insufficient physical activity by BMI area



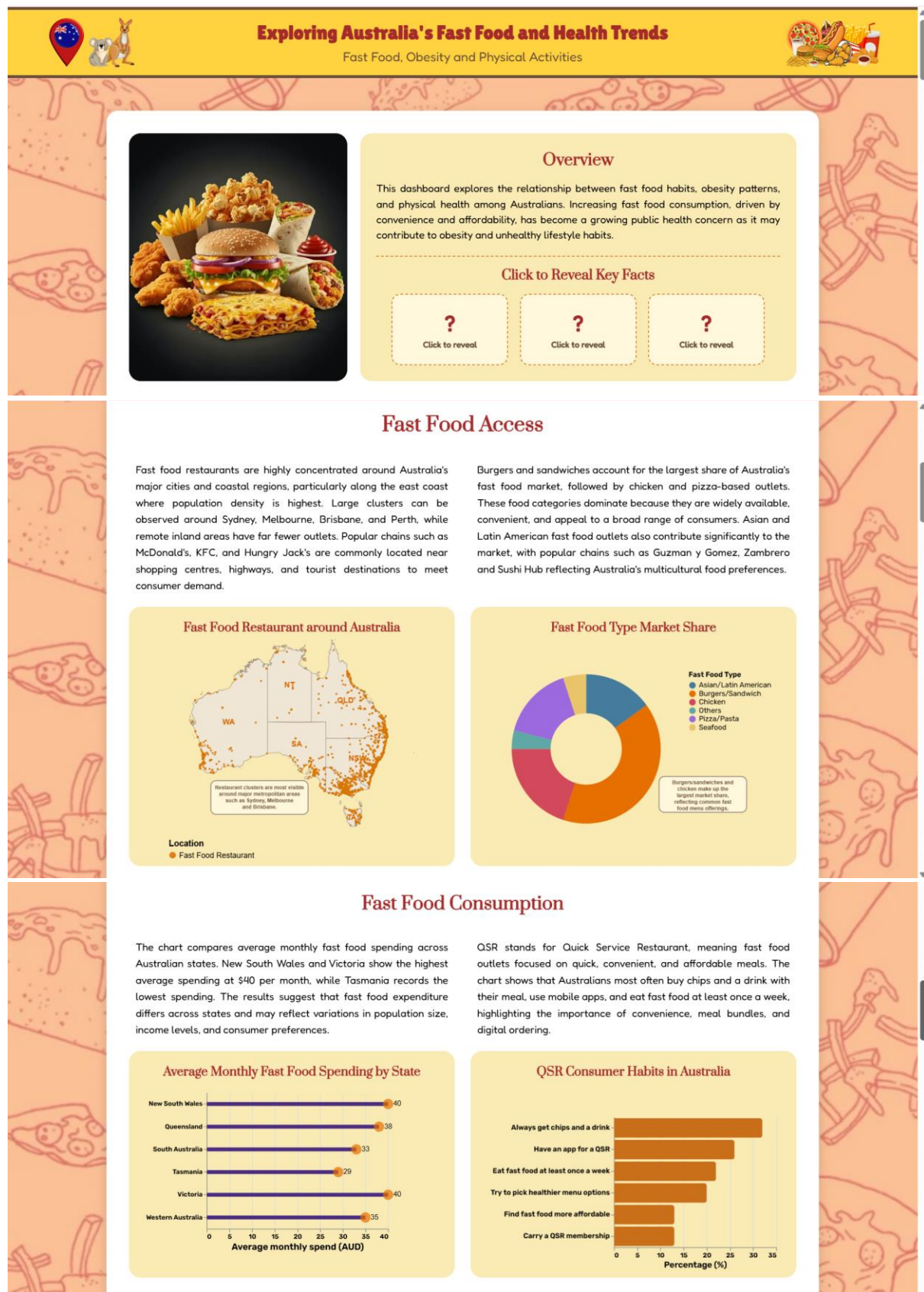
Physical activity completed last week in 2022



Conclusion

References

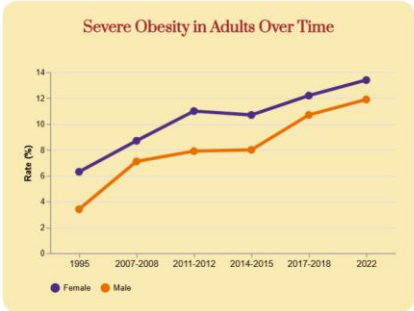
Part D: Full-size screenshot of your DV II



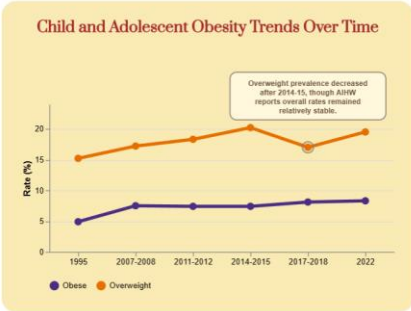
Fast Food and Obesity



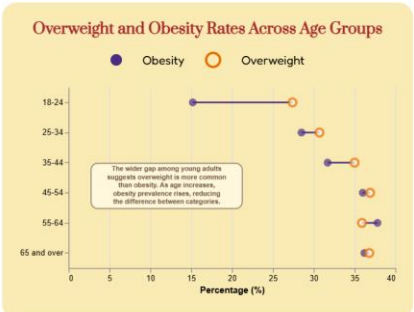
Increasing fast food accessibility and consumption patterns may contribute to rising obesity rates and unhealthy lifestyle habits across Australia. Higher fast food consumption is often associated with diets high in calories, fat, and sugar, which may increase the risk of obesity and related health issues. Obesity statistics used in this dashboard were obtained from the Australian Institute of Health and Welfare (AIHW).



Severe obesity among adults has increased steadily since 1995, with females consistently recording slightly higher rates than males. Across age groups, overweight and obesity are most common among middle-aged adults (45-64 years), while younger adults aged 18-24 show the lowest prevalence.



Although overweight prevalence declined between 2014-15 and 2017-18, AIHW reports that changes across this period were not statistically significant, suggesting the decrease may reflect normal survey variation rather than a substantial population-level improvement



Young adults have a bigger difference between overweight and obesity rates, meaning many are above a healthy weight but not yet obese. As people get older, this gap gets smaller because obesity becomes more common. This pattern may be due to long term habits like unhealthy eating, frequent fast food consumption, and not getting enough exercise.

Overall, these three charts show that fast food access, eating habits, and physical activity are strongly linked to obesity in Australia. The rise in overweight and obesity rates may be due to frequent fast food consumption and not enough exercise, highlighting the need for healthier lifestyle choices.

Body Mass Index (BMI)

BMI is commonly used to estimate whether a person is underweight, healthy weight, overweight, or obese based on height and weight. Higher BMI levels may be linked to increased risks of obesity, diabetes, heart disease, and other long-term health conditions.

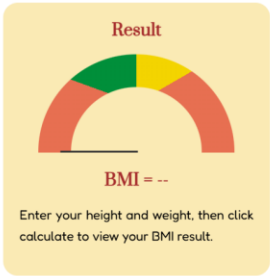
Adult BMI Calculator

Only calculates BMI for adults aged 20 and above.

Height (cm)

Weight (kg)

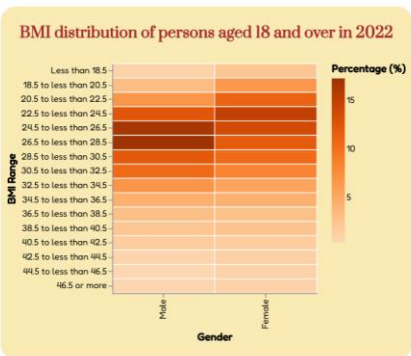
Calculate



BMI and Obesity

Women have a higher proportion of individuals within the normal BMI range, while men are more likely to be overweight. Obesity is prevalent in both genders, indicating that excess weight remains a significant health concern among Australian adults.

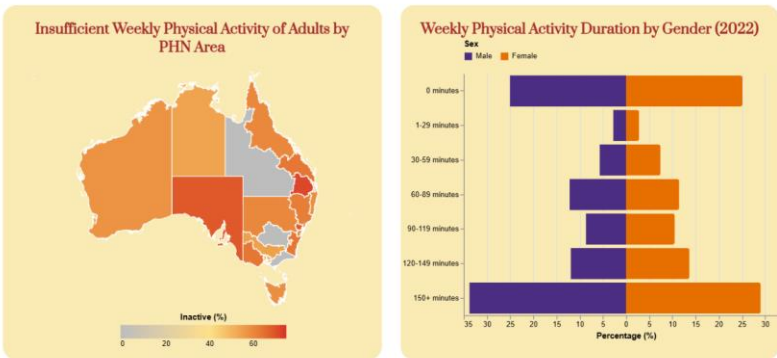
The heat map reveals that most Australian adults are concentrated within the overweight and obesity BMI ranges rather than the underweight categories. This trend is observed for both males and females, suggesting that excess weight remains a widespread public health issue across the adult population.



Obesity and Physical Activity

Primary Health Networks (PHNs) are healthcare planning regions established to coordinate health services based on local population needs. Each PHN is identified by a specific PHN area code, and the boundaries are defined according to healthcare service regions rather than state borders. The map shows that physical inactivity levels differ across PHNs, suggesting that factors such as access to facilities, population characteristics, and lifestyle behaviours may influence activity levels.

The chart shows the distribution of weekly physical activity among males and females. Although many adults achieved 150 or more minutes of activity, a considerable proportion reported no activity. Lower activity levels are associated with increased risks of overweight and obesity, highlighting the importance of regular exercise for maintaining good health.



Conclusion

Overall, this dashboard shows that fast food access, consumption habits, obesity patterns, BMI distribution, and physical activity are closely connected to public health outcomes in Australia. Areas with higher fast food availability and lower physical activity may face greater health risks, highlighting the importance of balanced diets, active lifestyles, and stronger health awareness.

Dashboard by Puteri Binti Ainul Hasni - 36853577

References

- pbin009. (2025). *Data-Visualisation-2* [Computer software] GitHub. [View Source](#)
- Australian Government. (n.d.). *Primary Health Network (PHN) locator map*. [View Source](#)
- Australian Institute of Health and Welfare. (n.d.). *Overweight and obesity data*. [View Source](#)
- Canstar Blue. (n.d.). *Cheapest fast food in Australia*. [View Source](#)
- GapMaps. (2025). *2025 Australia annual fast*

AI Acknowledgement

According to Monash University (2024), students must acknowledge the use of AI by following the guidelines provided on the Monash website.

I used AI in the following ways:

(i) generate ideas or structure suggestions, for assistance with understanding core concepts, or other substantial foundational and preparatory activity for the assessment.

AI was used to generate ideas for dashboard layout, chart selection, and storytelling structure. It was also used to improve understanding of data visualisation concepts and chart design choices.

(ii) generate text, rewrite, rephrase and/or paraphrase a portion of this assessment.

AI was used to refine and paraphrase chart descriptions, annotations, and dashboard narratives. All generated text was reviewed and edited to ensure it accurately reflected the data and project findings.

(iii) generate some other aspect of the submitted assessment.

AI was used to assist with HTML, CSS, and Vega-Lite coding by suggesting implementation approaches and debugging issues. All code was reviewed, modified, tested, and integrated into the final dashboard by me.